

Education

University of Illinois Urbana-Champaign	Champaign, USA
Master of Computer Science in Computer Science and Engineering	August 2026 - Decemeber 2027
RV College of Engineering	Bengaluru, India
Bachelor in Computer Science and Engineering (GPA: 9.22/10.0)	August 2019 - May 2023

Work Experience

Qualcomm India Private Limited	Bengaluru, India
Software Engineer	January 2023 - July 2026

- Designed and delivered a production-ready SoC upgrade framework under tight deadlines and limited testing resources
- Optimized the data-relay mechanism, increasing Bluetooth throughput by 24% and reducing system load during upgrades
- Refactored core component of DFU module, delivering two major client releases on schedule and reducing integration defects while improving system robustness
- Introduced GenAI-based tools for log-analysis automation which reduced triage time, and creating code documentation

Epsilon Pvt Ltd	Bengaluru, India
Software Engineering Intern	August 2022 - January 2023

- Benchmarked HBase, Apache Phoenix and Solr for large-scale data retrieval and analytics, evaluating filtering, aggregation, and multi-condition queries across varying dataset sizes.
- Optimized database query performance through primary//secondary indexing, query restructuring, and selective column retrieval, reducing representative Phoenix query execution from 19.4s to 0.37s.
- Built and evaluated a distributed data-processing environment using Hadoop/HDFS, HBase, Phoenix, Solr, and Docker, analyzing performance across storage and query architectures and documented benchmarking results for large datasets.

Project Experience

Business Documents Data Recognition and Table Extraction using Deep Learning Techniques

- Integrated DETR-based table extraction and YoloV5 object detection, achieving mAP values of 84.36% and 80.92% across diverse business documents
- Presented at International Conference of Artificial Intelligence and Computing (ICAIC) 2025. [Paper Link](#)

Novel Biomarker Prediction for Lung Cancer Using Random Forest Classifier

- Evaluated multiple ML and ensemble algorithms on gene expression analysis data to detect NSCLC/SCLC causing genes, and Random Forest predict potential biomarkers. Optimized the model to achieve 0.93 AUC and 87% accuracy
- Published in Cancer Informatics. [Paper Link](#)

Conditional Independence Testing using RCoT

- Modified RCoT algorithm to run on high-performance computing systems (HPCC), reducing runtime by 50% using synthetic data
- Published in JAIT Journal. [Paper Link](#)

Skills

Programming Languages: Python, Java, C/C++, SQL, Dart, R, Rust
Frameworks: ReactJS, NodeJS, Kotlin, Tensorflow, OpenSSL, Redis, PostgreSQL, SQLite, MongoDB
Tools: Jira, Git/Github, GCP, AWS, Shell Scripting, Jenkins, Perforce, Kubernetes, Terraform, Ansible
Machine Learning: CNN, RNN, LSTM BERT, Transformers, PyTorch, Keras, TensorFlow, Spacy, Langchain, GAN, VAE, RAG